

DDR基础课程

第一节 - 初识DDR

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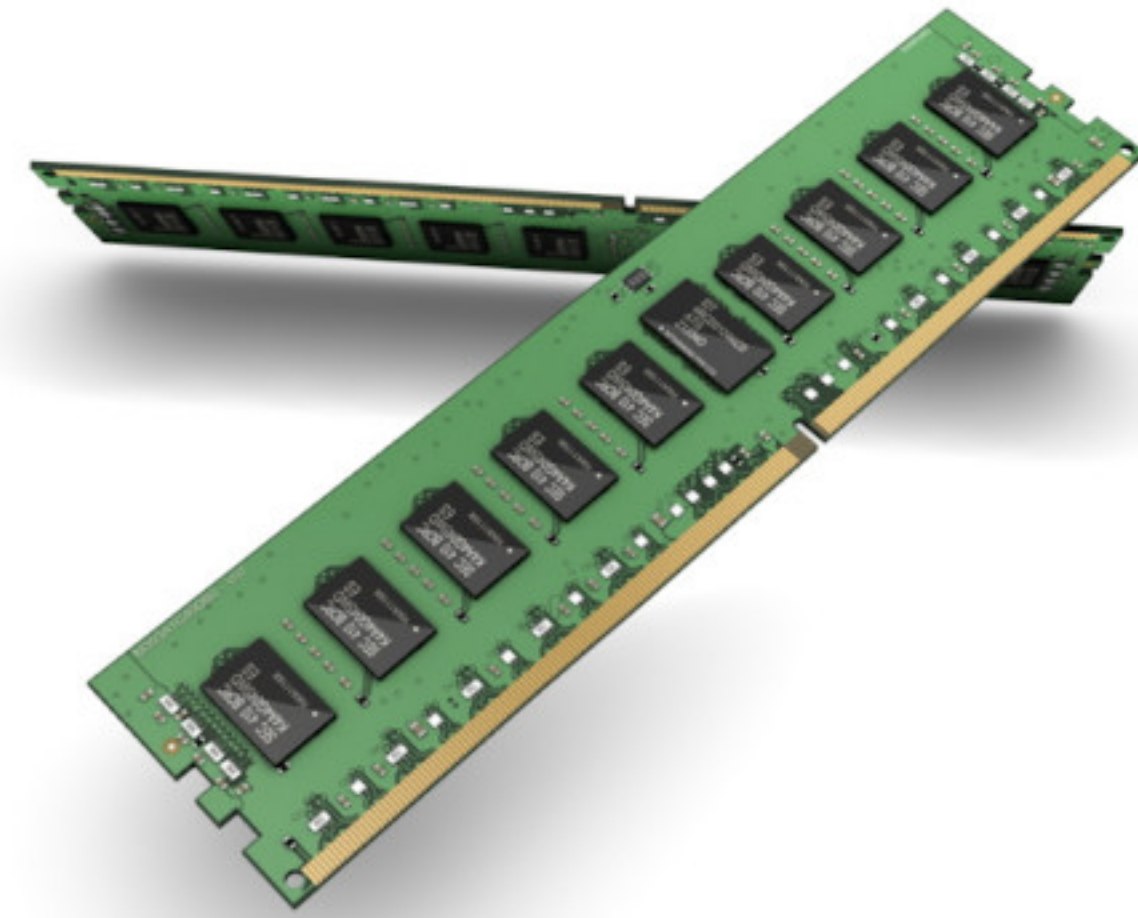
目录

关注的问题

- 什么是DDR?
- DDR的形态?
- 为什么要有DDR?
- DDR在计算结构中的作用?
- 为什么用DDR而不是其他解决方案?
- DDR的发展情况以及未来前景?

什么是DDR?

- DDR - Double Data Rate
 - 数据的速率是时钟速率的两倍
- DRAM - Dynamic Random Access Memory
- SoC DDR Subsystem - 用于访问DRAM的子系统模块



常见DRAM的形态

协议类型	主要使用类型	常见使用场合	特点	最新版本
DDR	DIMM	服务器 台式计算机	适用于对功耗不敏感的场所。存储密度高。方便更换。	DDR5
LPDDR	贴片	手机，手表，笔记本电脑	适用于移动设备等对于功耗敏感的场所。	LPDDR5X
GDDR	贴片	显卡，AI加速卡	主要用于显卡等这类超高速率要求，超高带宽要求的场合。	GDDR6X

常见DRAM的形态

DDR



XPS Desktop

★★★★★ 4.3 (1651) [Ask a question](#)

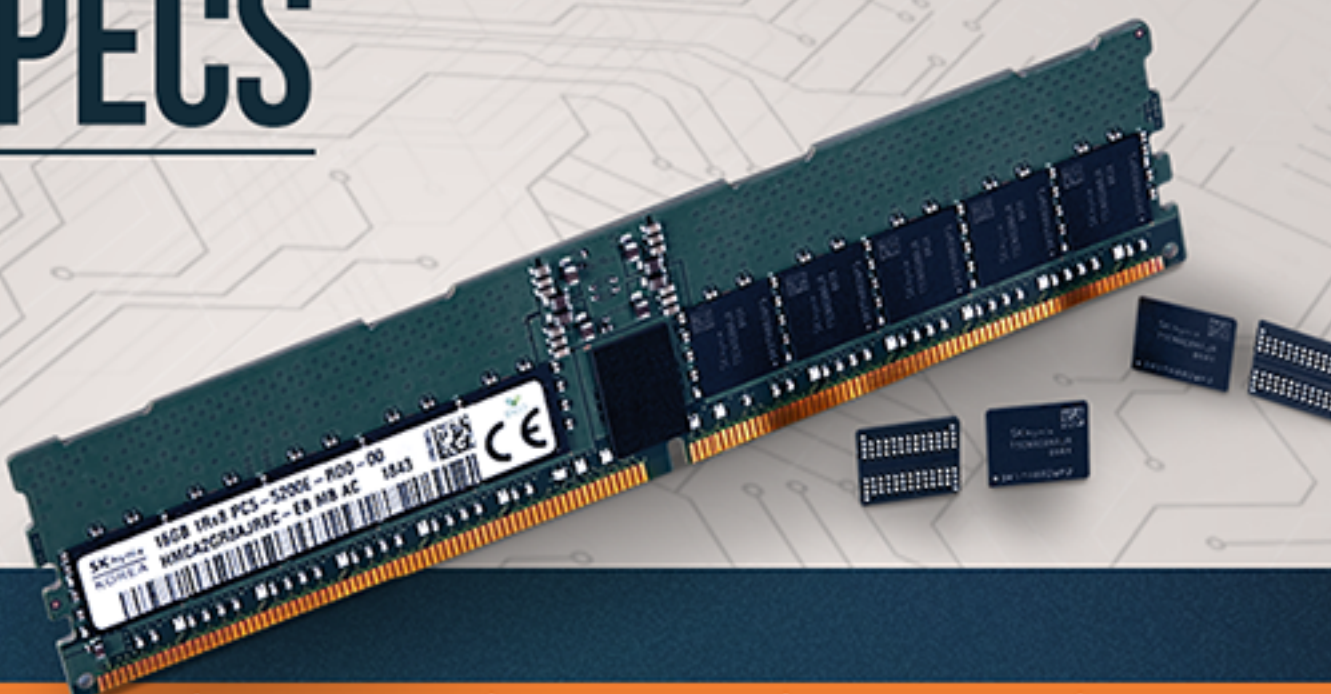
- 12th Gen Intel® Core™ i9-12900 (30 MB cache, 16.5 GHz Turbo)
- Windows 11 Home, English
- NVIDIA® GeForce RTX™ 3070, 8 GB GDDR6
- 32 GB, 2 x 16 GB, DDR5, 4400 MHz; up to 128 GB
- 1TB M.2 PCIe NVMe SSD (Boot) + 1TB 7200RPM S
- 750W Platinum Silver Bezel Chassis including opti
- [Ports & Slots](#)

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DRAM SPECS



	DDR5	DDR4	DDR3	DDR2	DDR
Data Transfer Rate (Mbps)	3200~6400	1600~3200	800~1600	400~800	200~400
Power Consumption	1.1V	1.2V	1.5V / 1.35V	1.8V	2.5V
Capacity	8Gb~32Gb	4Gb~16Gb	512Mb~4Gb	128Mb~2Gb	64Mb~1Gb

常见DRAM的形态

LPDDR

More Transistors, More Memory

The system-on-a-chip (SoC) design of M2 is built using enhanced, second-generation 5-nanometer technology, and consists of 20 billion transistors — 25 percent more than M1. The additional transistors improve features across the entire chip, including the memory controller that delivers 100GB/s of unified memory bandwidth — 50 percent more than M1. And with up to 24GB of fast unified memory, M2 can handle even larger and more complex workloads.

Mode	LPDDR5X	LPDDR5	LPDDR4(X)	LPDDR3	LPDDR2	LPDDR
B/W (Mbps)	7466~8532	5500~6400	3200~4266	1600~1866	800~1066	333~400
VDD/VDDQ*	1.05V/0.5V	1.1V/1.1V(0.6V)	1.2V/1.2V	1.2V/1.2V	1.2V/1.2V	1.2V/1.2V
Die Density	8Gb~24Gb	4Gb~24Gb	1Gb~8Gb	256Mb~4Gb	128Mb~2Gb	
Release Year	2023 (Expected)	2020	2014	2012	2010	2008

* VDDQ: Power supply for I/O (Input/Output); decreased VDDQ leads to less power consumption



Apple Silicon M2 CPU

Performance Cores	4
Efficiency Cores	4
Transistors	20 Billion
Core Architecture (P- / E-Core)	Avalanche / Blizzard (not confirmed)
Memory	24 GB of LPDDR5 (100 GB/s)
Share L2 Cache (P- / E-Core)	16 MB / 4 MB
Instruction Cache (P- / E-Core)	192 / 192 KB
Data Cache (P- / E-Core)	128 / 64 KB
Neural Engine	15.8 TOPS

常见DRAM的形态

GDDR

Table 1: Categories of Ultra-Bandwidth Memory

Ultra-Bandwidth Solutions Memory Comparison	GDDR5	GDDR5X	GDDR6	GDDR6X
	Graphics Card	Graphics Card	Graphics Card AI Inference Accelerator	Graphics Card AI Inference Accelerator
Application Type (Example)	GTX 1070 RX 580	TitanX	Titan RTX RX5700 XT	RTX 3090
# of placements	8	12	12	12
Gb/s/pin	8	11.4	14-16	19-21
GB/s/placement	32	45	56-64	76-84
GB/s/system	256	547	672-768	912-1008
Configuration (Example)	256 IO/s (8pcs x 32 IO package)	384 IO/s (12pcs x 32 IO Package)	384 IO/s (12pcs x 32 IO Package)	384 IO/s (12pcs x 32 IO Package)
Frame Buffer of Typical System	8GB	12GB	12GB	12GB
AVG Device Power (pJp)	9.0	8.0	7.5	7.25
Typical IO channel	PCB (P2P SM)	PCB (P2P SM)	PCB (P2P SM)	PCB (P2P SM)

Gigabyte GeForce RTX 3080, 10GB GDDR6X, Gaming OC rev. 2.0 (LHR)



Picture may differ from actual product, there can be accessories and parts not supplied with the actual product

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Delivery options

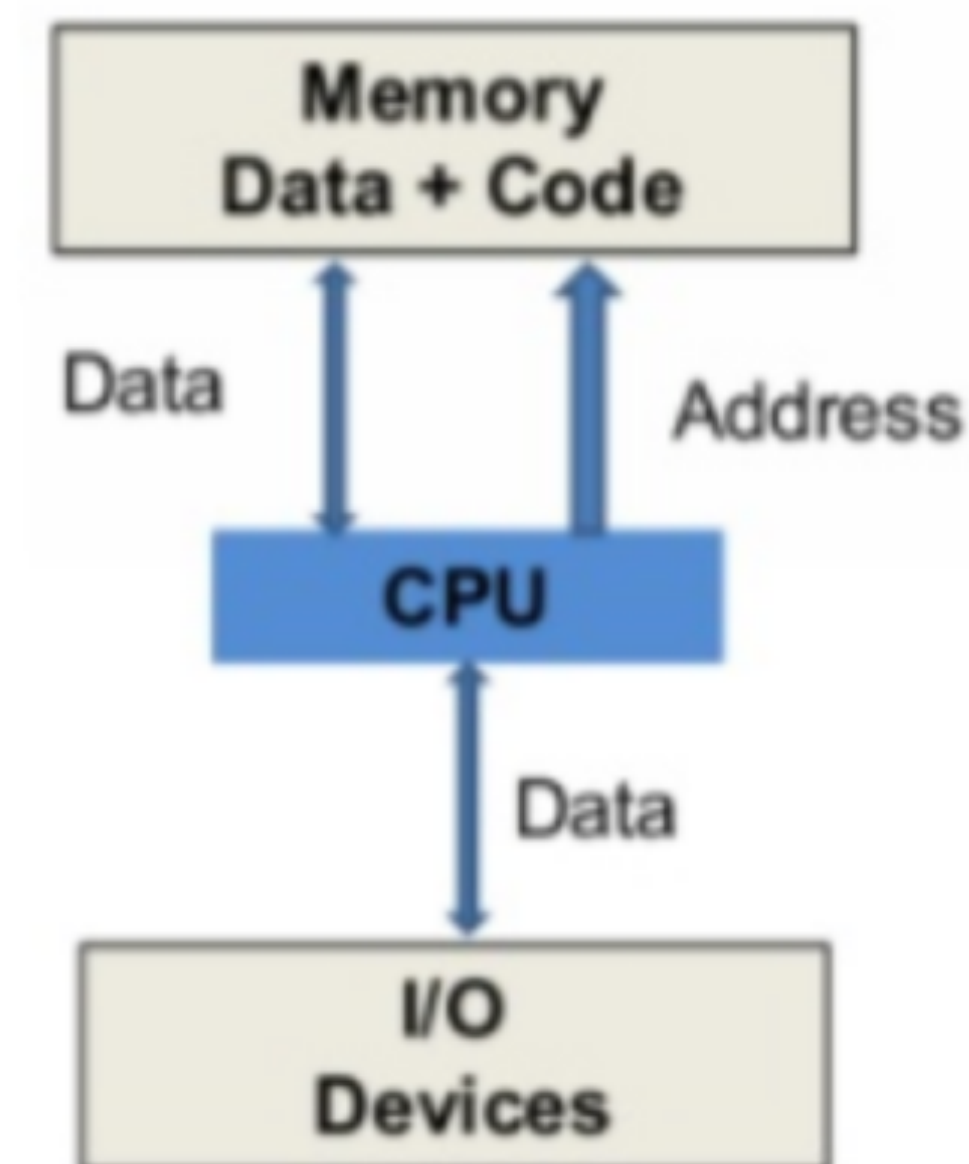
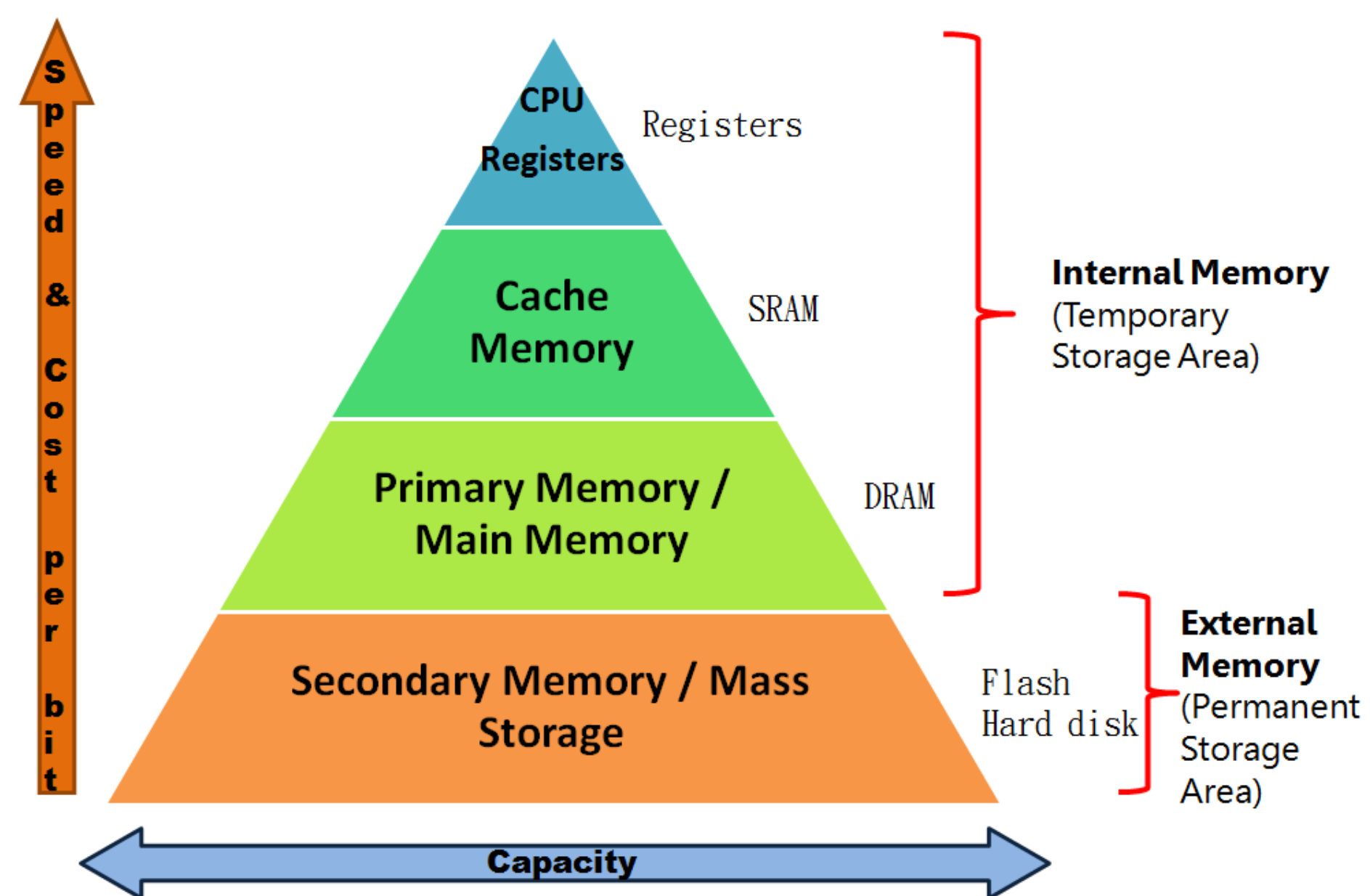
- Delivery to Omniva parcel machine free of charge: on 15.09.2022
- Piegāde uz DPD Pickup Paku Skapi free of charge: on 15.09.2022
- Delivery to address in Riga 3.99 €: on Thursday, 15.09.2022
- Delivery to address in Latvia 6.99 €: on Thursday, 15.09.2022
- Delivery to Dateks.lv parcel drop-off point at 9 Zemīte: on Tuesday, 13.09.2022 after 15:00



为什么要有DDR?

DDR在计算结构中的作用

- DDR的作用是存储，为什么一定要用DDR来作为存储？
- 计算架构 - 冯诺依曼架构
- 存储架构



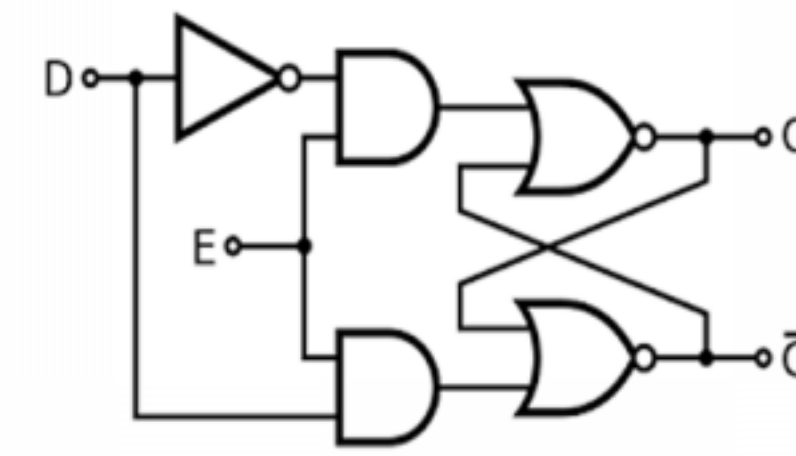
Von Neumann Machine

为什么要有DDR?

DDR在计算结构中的作用

- Register作为存储单元
- 响应速度很快
- 一般需要20个Transistor才能组成一个DFF
- 面积很大，非常昂贵

D Flip Flop Circuit HEF4013B



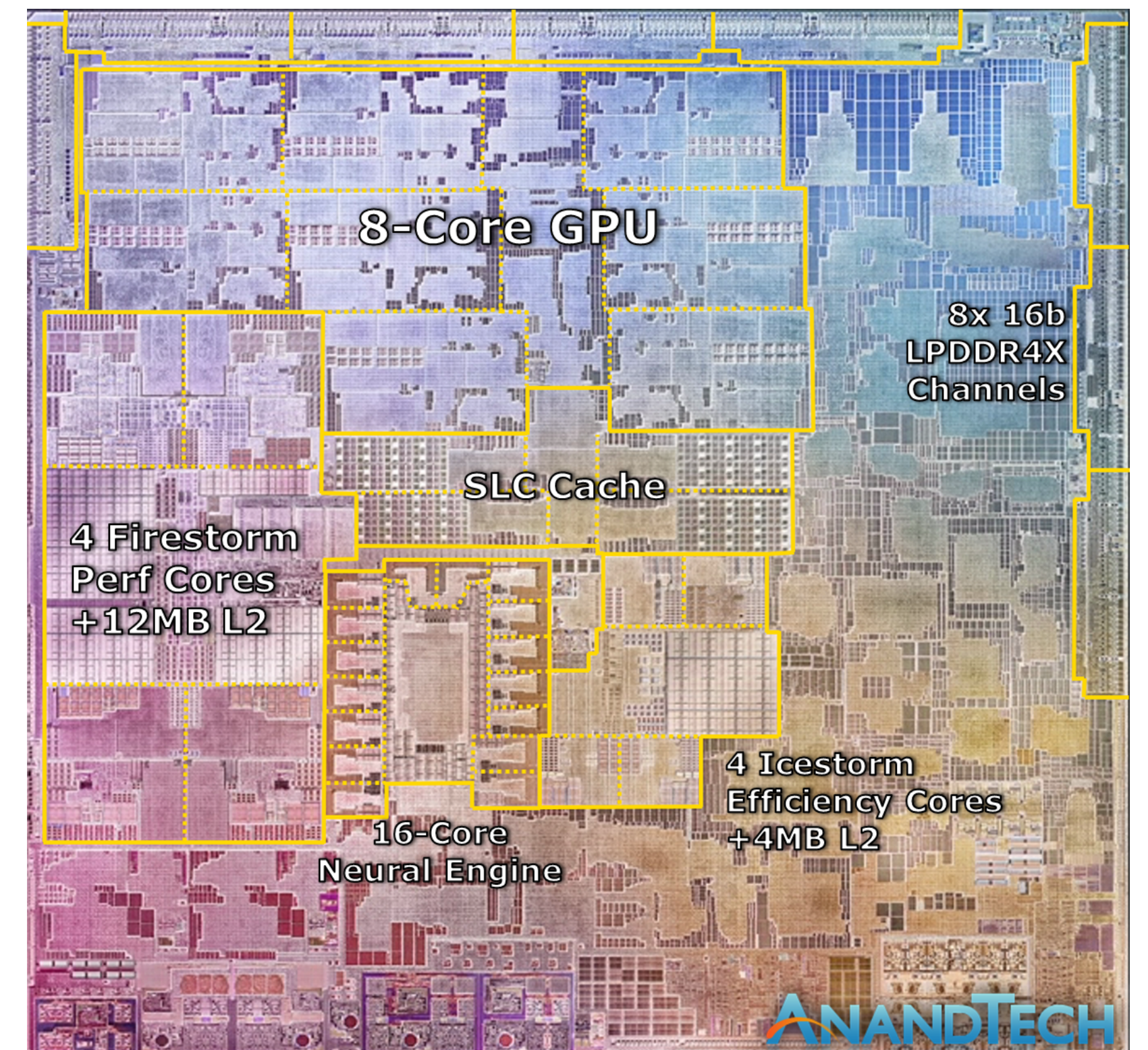
D	S	R	Q	State
	0	0	Previous State	No Change
0	0	1	0	Reset
1	1	0	1	Set
	1	1	?	Forbidden

SR & D Flip Flop TruthTable



DDR在计算结构中的作用

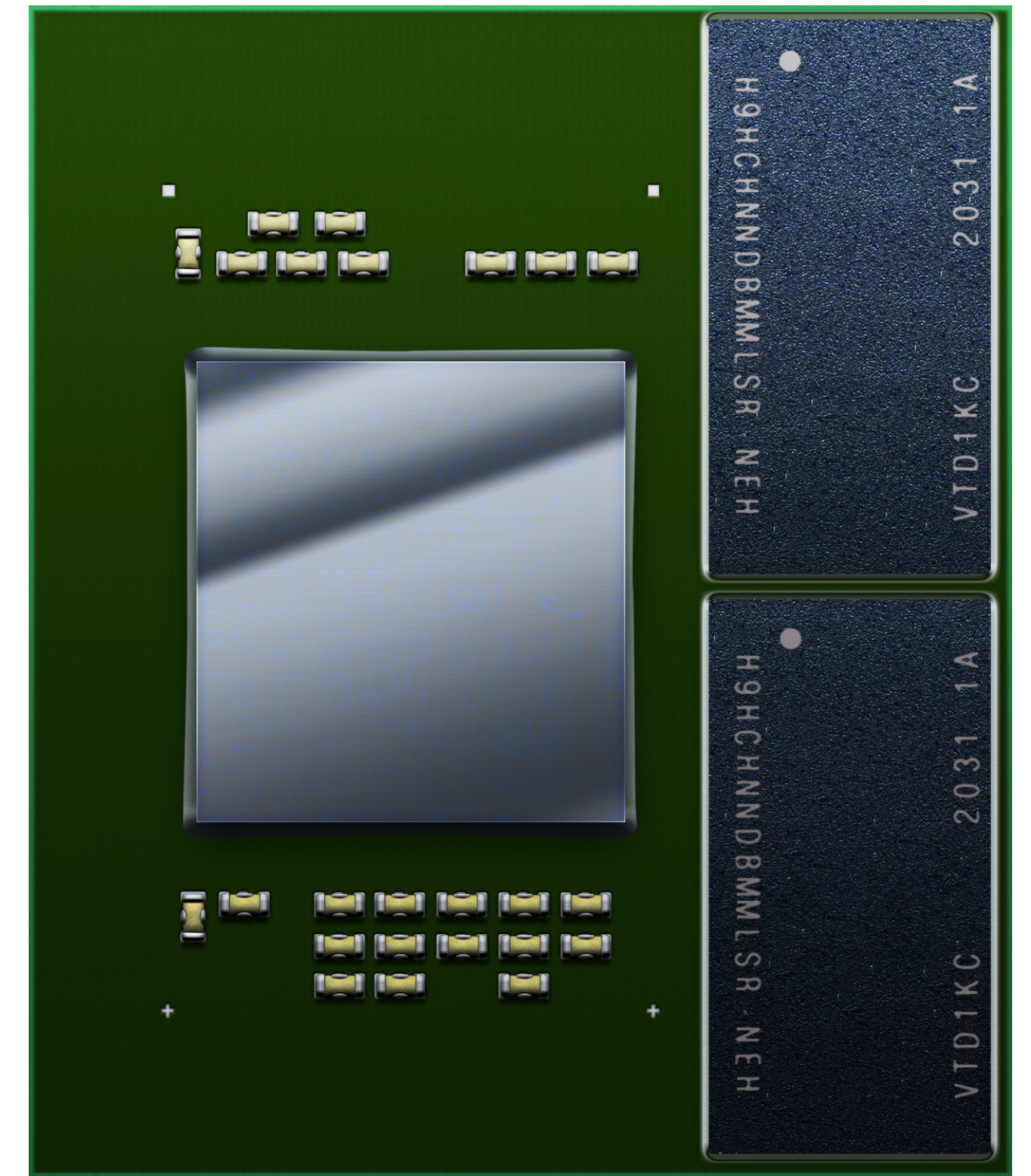
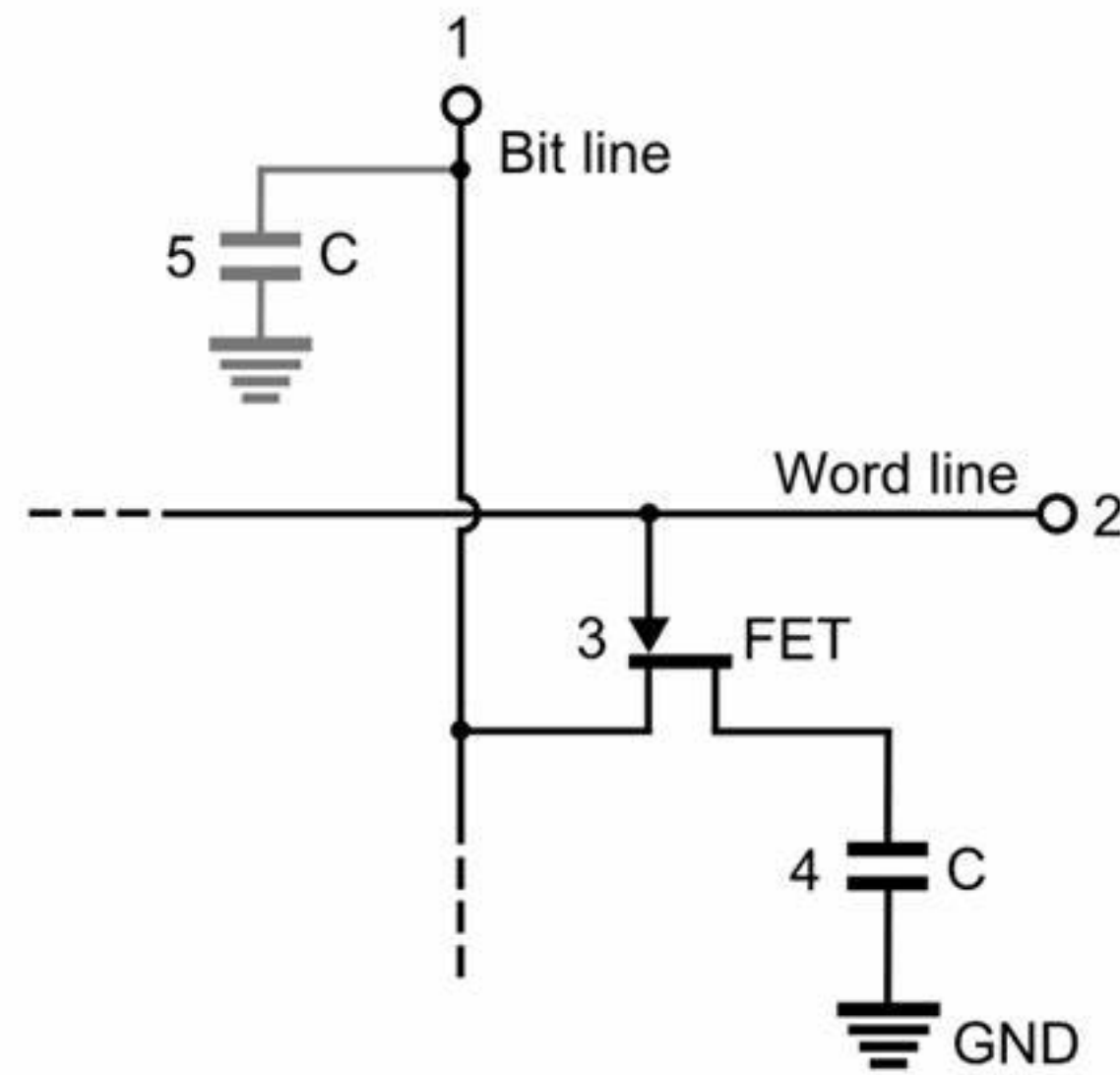
- | | |
|-------------------------|-----------------|
| Last level cache | M1: 8 MB |
|-------------------------|-----------------|



为什么要有DDR?

DDR在计算结构中的作用

- DRAM作为存储单元
- 工艺与逻辑工艺不同
- 一个Transistor及一个Capacitor
- Transistor作为Switch, Capacitor作为存储
- 面积非常小, 价格大幅度降低
- 缺点: 片外, 响应速度慢



The SoC has access to 16GB of unified memory.

为什么要有DDR?

DDR在计算结构中的作用

- Flash/硬盘作为存储单元
- 工艺与逻辑工艺不同
- 价格大幅度降低
- 缺点： 响应速度非常慢，无法支持实时计算

DDR的未来

- DDR在未来的很长一段时间里依旧会是存储的主要角色
- DDR在未来会继续沿着更高速率的目标发展
- 低功耗重要性越发突出

DDR6 and GDDR6 are ideal options as the demand for universal memory devices and enterprise storage applications increases. There will continue to be a huge market demand for DDR6 RAM due to its high bandwidth, low latency, and low power consumption. Technologies such as the [Internet of Things](#) (IoT), big data, and artificial intelligence (AI) will continue to use DDR technology.

DDR6 RAM: Advantages and Challenges

 CADENCE PCB SOLUTIONS

Key Takeaways

- DDR6 RAM is the latest among the DDR iterations and the maximum data rate peaks above 12000 MT/s.
- The conductor and dielectric materials used in DDR6 memory interfaces influence signal integrity at high data rates.

During the event, Samsung also talked about its new iteration of GDDR6 memory based on the company's 12nm process called GDDR6+. This new variant of GDDR6 memory will increase its maximum speed from 18,000Mbit/s to 24,000Mbit/s, which translates into a 33% increase over its predecessor. Moreover, GDDR6+ will be succeeded by GDDR7, which is said to feature real-time error protection capabilities and a maximum data rate of 32,000Mbit/s.

When talking about LPDDR6 memory, Samsung stated its focus was on improving power efficiency, but speed will still improve considerably. Samsung plans to increase efficiency by about 20% and increase data rates up to 17,000Mbit/s. Finally, Samsung also talked about its plans to manufacture HBM3 with 800GB/s bandwidth and HBM2E with 450GB/s bandwidth. Samsung plans to begin production of its HBM3 memory in Q2 2022.

下节预告

第二节 - DRAM的特点

- DRAM相比于SRAM的关键区别
- DRAM的特点带来的设计挑战
- DRAM的刷新

谢谢